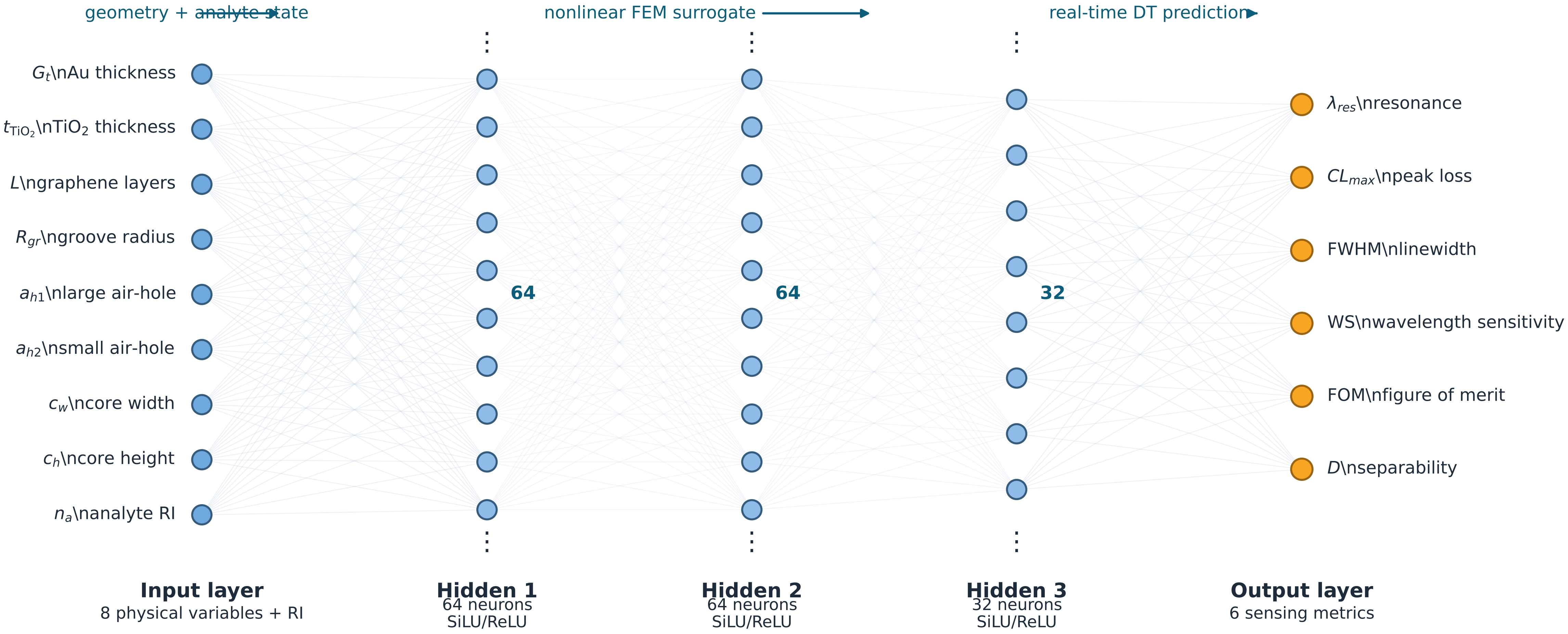
# Surrogate architecture for the SPR-PCF digital twin

FEM-trained target model: 9 inputs → 64 → 64 → 32 → 6 outputs




Digital-twin role: maps the eight SPR-PCF physical design variables and analyte refractive index to resonance-response metrics. The current interface may use a demonstrator surrogate; publication-grade prediction requires retraining on geometry-level FEM/COMSOL data, uncertainty calibration, and independent FEM verification.